

HOMEWORK 3, DUE SEPTEMBER 24

(10 points) 1. Prove that for any $x, y \in \mathbb{R}$,

$$||x| - |y|| \leq |x - y|.$$

(10 points) 2. Use the limit definition to compute

$$\lim_{n \rightarrow \infty} \frac{n+2}{3n+5}.$$

(10 points) 3. Prove that if

$$\lim_{n \rightarrow \infty} x_n = \infty$$

and $(y_n)_n$ is bounded, then

$$\lim_{n \rightarrow \infty} (x_n + y_n) = \infty$$

(10 points) 4. Let $(x_n)_n$ be an increasing sequence of positive real numbers. Define

$$y_n = \frac{x_1 + x_2 + \dots + x_n}{n}.$$

Prove that (y_n) is increasing as well.